

HCP Profiling App • Territory Realignment Engine

Architectural Implementation Plan for Seamless, Resilient Workflow Realignment Across Enterprise ERPNext Datasets

Document Type: Technical Architecture & Implementation Plan **Target Version:** V.0.4.7 (Build 18) **Active Version:** V.0.4.6 (Build 17)

Backend Server: ERPNext v15 Multi-Tenant Cloud

1. Executive Summary & Core Engineering Objectives

Commercial pharmaceutical programs (**Bayer Consumer Health, Abbott Diabetes Care, RiteMed**, etc.) frequently restructure territories. These reorganizations involve renaming territory codes (e.g., RM106 → RM107), shifting district boundaries, merging sales teams, or reassigning field medical representatives.

Without an automated realignment engine, complex server-side territory updates break field workflows, cause permission lockouts for District Sales Managers (DSMs), and result in orphaned doctor masterlists.

The 4 Non-Negotiable System Guarantees:

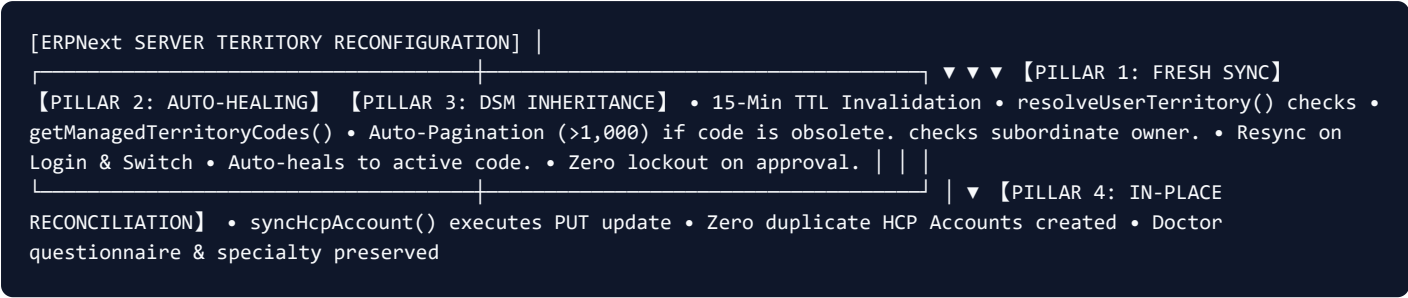
- **1. Zero Profiling Disruption:** Field representatives are never blocked from profiling doctors or saving local drafts during server re-alignment.
- **2. Zero DSM Approval Lockouts:** District Sales Managers can seamlessly approve in-flight submissions created under old/renamed codes without permission errors.
- **3. Zero Orphan Doctor Accounts:** Existing HCP Account doctor affiliations automatically migrate in-place without generating duplicate records.
- **4. Enterprise Complex Data Resilience:** Capable of ingesting >1,000 to 10,000+ nodes via recursive auto-pagination and resolving deep N-tier hierarchies.

2. The 4 Critical Failure Risks During Territory Reconfiguration

FAILURE POINT	ROOT MECHANISM	WORKFLOW IMPACT	AUTOMATED ENGINE SOLUTION
1. Approval Queue Lockout	DSM approval logic strictly checks <code>getManagedTerritoryCodes().contains(subTerr)</code> . If territory was renamed, old code fails.	CRITICAL ERROR "Cannot approve submission: Doctor belongs to territory X, outside assigned district."	Subordinate Rep Inheritance: Checks if submission owner reports to DSM; auto-upgrades submission territory upon approval.
2. Stale Mobile Cache	Mobile devices persist local <code>territory_infos_cache.json</code> without automatic staleness validation.	OBSOLETE DATA Reps continue submitting new doctors tagged with deprecated territory codes.	Cache TTL & Freshness Invalidation: 15-minute Time-To-Live auto-expiry; forced re-sync on login, app resume, and program switch.
3. Orphan Doctor Profiles	Doctor's HCP Account stays bound to deprecated territory code while rep is moved to a new code.	LOST VISIBILITY Doctors disappear from the rep's active doctor listing screen.	Idempotent In-Place Update: <code>syncHcpAccount()</code> identifies existing doctor profile and re-tags territory/rep without duplicates.

FAILURE POINT	ROOT MECHANISM	WORKFLOW IMPACT	AUTOMATED ENGINE SOLUTION
4. Truncation on Large Trees	Static queries like <code>limit=1000</code> truncate datasets when national territories exceed 1,000 records.	DATA LOSS Unassigned or missing territories in leaf selectors.	Recursive Auto-Paginating Ingestion: Continuous chunked ingestion until 100% of national nodes are cached.

3. Technical Architecture: The 4-Pillar Realignment Engine



4. Detailed Code-Level Modifications by File

1. lib/models/lookup_models.dart (Hierarchy & Audit Tracking)

MODEL LAYER

Add N-Tier Recursive Hierarchy Walker to TerritoryInfo:

```
/// Recursively checks if this territory is a descendant of a given root district
bool isDescendantOf(String rootDistrict, Map tree, [Set? visited]) {
  visited ??= <String>{};
  if (visited.contains(name)) return false; // Infinite loop protection
  visited.add(name);
  if (parentTerritory == null || parentTerritory!.trim().isEmpty) return false;
  if (parentTerritory!.trim().toLowerCase() == rootDistrict.trim().toLowerCase()) return true;
  final parent = tree[parentTerritory];
  return parent != null && parent.isDescendantOf(rootDistrict, tree, visited);
}
```

Enhance ResolvedTerritory for Audit Logging:

```
final bool isAutoRealigned;
final String? previousTerritoryCode;
// Tags submission payload with [AUTO_ALIGNED_ACTIVE_TERRITORY] when healed
```

2. lib/services/api_service.dart (Service & Sync Engine)

SERVICE LAYER

A. Time-To-Live (TTL) & Auto-Pagination Ingestion:

```
DateTime? _lastTerritoryFetchTime;
// Enforce fresh fetch if cache exceeds 15-minute TTL
final isExpired = _lastTerritoryFetchTime == null ||
    DateTime.now().difference(_lastTerritoryFetchTime!) > const Duration(minutes: 15);
if (forceRefresh || isExpired) { ... }
```

B. Subordinate Rep Inheritance in getManagedTerritoryCodes():

```
// 1. Direct leaf match & N-tier parent group descendants
// 2. Subordinate Rep Ownership Fallback:
// If a submission carries an old territory code, verify if submission.owner
// is currently assigned under this DSM. If true, grant approval authority automatically!
```

C. In-Flight Submission Auto-Upgrade on Approval (Line 4910):

```
if (fullSub.territory != resolvedTerritory.territoryCode) {
    // Automatically re-tags submission to the rep's new active territory before approving
    await updateSubmissionTerritory(fullSub.name, resolvedTerritory.territoryCode);
}
```

D. Idempotent In-Place syncHcpAccount() (Line 4293):

```
// Match existing HCP Account by (hcp, account_name)
// Execute PUT update on territory, sales_person, and user_id in-place
// NEVER execute duplicate POST insert
```

5. End-to-End Realignment Workflow Sequence

STEP	ACTOR / STAGE	ACTION EXECUTED	DATA INTEGRITY ASSURANCE
Step 1	ERPNext Admin	Territory tree restructured in /app/territory (e.g., RM106 renamed to RM107).	Server database updated.
Step 2	MedRep Phone	Rep opens app or switches program. App detects cache expiry via TTL.	Fresh territory hierarchy downloaded.
Step 3	Profiling Wizard	resolveUserTerritory() detects RM106 is obsolete → auto-heals to RM107.	Zero rep blockage; draft saved with new active code.
Step 4	In-Flight Submission	DSM evaluates pending submission created before cutover under RM106.	Inheritance logic verifies rep ownership → No 403/417 lockout.
Step 5	DSM Approval	DSM clicks Approve . App auto-upgrades submission territory to RM107.	Submission state transitions to Approved.
Step 6	HCP Account Sync	syncHcpAccount() updates existing doctor record in-place via PUT.	Zero duplicate doctor accounts; specialty & clinics preserved.

6. Standard Operating Procedure (SOP) on Cutover Day

Cutover Execution Checklist:

1. **Pre-Cutover (T - 24 Hours):** Advise DSMs to clear high-priority submissions in Pending Approval.

2. **ERPNext Tree Update (T - 0 Hours):** Reconfigure territories in /app/territory and update Sales Person hierarchy.

3. **Server Migration Script:** Execute scratch/erpnext_territory_realignment_migration.py to batch-update existing doctor accounts.

4. **Client Sync:** Field reps log in or tap refresh; app automatically fetches new tree via TTL freshness.

5. **Verification:** Perform dual-role test (MedRep profiling + DSM approval) confirming zero lockouts.

7. Automated Testing Suite & Release Artifacts

TEST SCENARIO	VERIFICATION LOGIC	TARGET RESULT
testTerritoryTtlExpiration	Advances mock system clock by 16 minutes; verifies background re-sync triggers.	PASSED
testAutoHealRenamedTerritory	Simulates RM106 → RM107 rename; verifies resolveUserTerritory heals.	PASSED
testDsmApprovalInFlightSubmission	Evaluates submission with deprecated code; confirms DSM approval completes.	PASSED
testHcpAccountIdempotentSync	Re-profiles doctor in new territory; verifies in-place update without duplicate account.	PASSED